STAT 545 HW 8

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Problem 1

(Probit Regression). Consider the probit regression model for n independent observations $\{X_i Y_i\}$ where $Y_i \in \{0, 1\}$, $X_i \in R^p$ and we have $Pr(Y_i = 1) = \Phi(\beta^T X_i)$, for p-dimensional parameter vector , where $\Phi(\cdot)$ denotes the standard normal cdf. Suppose, we decide to put a at prior on β, i.e. $\pi(\beta) \propto 1$.

A.

Write down the posterior $\pi(\beta|X, Y)$ up to a constant of proportionality.

Each observation can either be 0 or 1 (Bernoulli) so the pmf for a single observation with given X and Beta is:

$$\Pr(Y_i|X_i, \beta) = [\Phi(\beta^T X_i)]^{Y_i}[1 - \Phi(\beta^T X_i)]^{1-Y_i}$$

We can combine them all together to get the joint likelihood function:

$$L(Y|\beta, X) = \prod_{i=1}^{n} \Phi(\beta^T X_i)^{Y_i}(1 - \Phi(\beta^T X_i))^{1-Y_i}$$

We take the non-infromative prior: $\pi(\beta) \propto 1$

Now we can compute the posterior:

$$\pi(\beta|X, Y) \propto L(\beta, Y|X) \propto L(Y|X, \beta) \cdot \pi(\beta)$$

Which is just:

$$\pi(\beta|X, Y) \propto \left(\prod_{i=1}^{n} \Phi(\beta^T X_i)^{Y_i}(1 - \Phi(\beta^T X_i))^{1-Y_i} \right) \cdot 1 = \prod_{i=1}^{n} \Phi(\beta^T X_i)^{Y_i}(1 - \Phi(\beta^T X_i))^{1-Y_i}$$

B.

Introduce the latent variable $Z_i = \beta^T X_i + \epsilon_i$ where ϵ_i are independent standard Gaussian. Show that β is conditionally independent of Y given Z and X. In particular, show that $\beta|X, Y, Z \sim Normal((X^T X)^{-1} X^T Z, (X^T X)^{-1})$

Let $Z_i = \beta^T X_i + \epsilon_i$ and $\epsilon_i \sim N(0,1)$.

We start with the conditional distribution of beta and Y and find:

$$\pi(\beta, Y \mid Z, X) = \pi(\beta \mid Z, X) \cdot \pi(Y \mid \beta, Z, X)$$

We know that

$$Y_i = \begin{cases} 1 & \text{if } Z_i > 0 \\ 0 & \text{if } Z_i \leq 0 \end{cases}$$

Then,

$$L(Y|Z, \beta, X) = \prod_{i=1}^{n} \Phi(Z_i)^{Y_i}(1 - \Phi(Z_i))^{1-Y_i}$$

which is not dependent on the X or β. Then, $\pi(Y \mid \beta, Z, X) = \pi(Y \mid Z)$.

$$\pi(\beta, Y \mid Z, X) = \pi(\beta \mid Z, X) \cdot \pi(Y \mid Z)$$

Because the distribution of Y given Z does not involve β and the distribution of β does not involve Y, we conclude that β and Y are conditionally independent given Z and X.

Then, we solve for the distribution of β (remember that $Z = X\beta + \epsilon$ is Normal$(X\beta, I_n)$:

$$\pi(\beta \mid X, Z) \propto f(Z \mid X, \beta)\pi(\beta) \propto \exp\left(-\frac{1}{2}(Z - X\beta)^T(Z - X\beta)\right) * 1$$

We expand the quadratic part and find:

$$(Z - X\beta)^T(Z - X\beta) = Z^T Z - 2\beta^T X^T Z + \beta^T X^T X\beta$$

Let $\Sigma^{-1} = X^T X \implies \Sigma = (X^T X)^{-1}$ and then $\Sigma^{-1}\mu = X^T Z \implies \mu = (X^T X)^{-1} X^T Z$. We find:

$$\pi(\beta \mid X, Z) \propto \exp\left(-\frac{1}{2}[\beta^T(X^T X)\beta - 2\beta^T(X^T Z)]\right) \propto \exp\left(-\frac{1}{2}(\beta - \mu)^T \Sigma^{-1}(\beta - \mu)\right)$$

Which tells us that:

$$\beta \mid X, Y, Z \sim Normal((X^T X)^{-1} X^T Z, (X^T X)^{-1})$$

C.

Show that

$$Z_i | X_i, Y_i, \boldsymbol{\beta} \sim \begin{cases} \mathrm{TN}_-(\boldsymbol{\beta}^{\mathrm{T}} X_i, 1, 0) & \text{if} \quad Y_i = 0 \\ \mathrm{TN}_+(\boldsymbol{\beta}^{\mathrm{T}} X_i, 1, 0) & \text{if} \quad Y_i = 1 \end{cases}$$

> where $TN_-(\mu, sigma^2, c)$ denotes a normal random variable with mean μ, variance σ^2, truncated from the right at c and similarly $TN_+(\mu, sigma^2, c)$ denotes a normal random variable with mean μ, variance σ^2, truncated from the left at c.

Using Bayes equation, we know:

$$f(z_i \mid X_i, Y_i, \beta) \propto f(Y_i \mid z_i, X_i, \beta) f(z_i \mid X_i, \beta)$$

We are given from the previous section that $Z_i = \beta^T X_i + \epsilon_i$ with $\epsilon_i \sim N(0,1)$, so $Z_i \mid X_i, \beta \sim N(\beta^T X_i, 1)$.

Next, we know that $f(Y_i \mid z_i, X_i, \beta) = f(Y_i \mid z_i)$, so Y is determined by Z. Then, $f(Y_i \mid z_i) = \mathbb{I}(z_i > 0)$ if $Y_i = 1, \quad \text{and} \quad \mathbb{I}(z_i \leq 0)$ if $Y_i = 0$.

Ok, now lets look at the cases like in the problem statement.

When $Y_i = 1$, $f(z_i \mid X_i, Y_i = 1, \beta) \propto f(z_i \mid X_i, \beta) \cdot \mathbb{I}(z_i > 0)$, which is just the regular normal density truncated on the left at $c = 0$.

When $Y_i = 0$, $f(z_i \mid X_i, Y_i = 0, \beta) \propto f(z_i \mid X_i, \beta) \cdot \mathbb{I}(z_i \leq 0)$, which is just the regular normal density truncated on the right at $c = 0$.

We combine the two cases and we have the identity as requested.

D.

> Take the true $\beta = (-3, 0, 3, -2, 4)^T$. Generate 500 inpendent $X_i = Normal(0, I_5)$ and simulate Y_i with probability of success $\phi(X_i^T \beta)$. Describe the steps in a data augmentation scheme to generate samples from $\pi(\beta|X, Y)$. Implement the scheme in R. Run the data augmentation scheme for 10,000 iterations with 5,000 burn-in steps. Plot a histogram of the posterior samples for each component of β.

We want to create a Gibbs sampler. We are given X and Y. We need to sample Beta and Z. The steps are:

1. Start with an initial guess $\beta^{(0)}$.
2. Sample Latent Variables $(Z \mid \beta, X, Y)$: For each $i = 1, \ldots, n$, sample:
    1. If $Y_i = 1$, $Z_i \sim \text{TN}_+(X_i^T \beta, 1, 0)$ (Normal truncated below at 0).
    2. If $Y_i = 0$, $Z_i \sim \text{TN}_-(X_i^T \beta, 1, 0)$ (Normal truncated above at 0).
3. Sample Parameters $(\beta \mid Z, X)$: Treating Z as the response in a linear regression:
    - $\beta \sim \text{Normal}\left((X^T X)^{-1} X^T Z, (X^T X)^{-1}\right)$.
4. Repeat: Iterate steps 2 and 3 for the desired number of samples that we need (10,000 in our case).

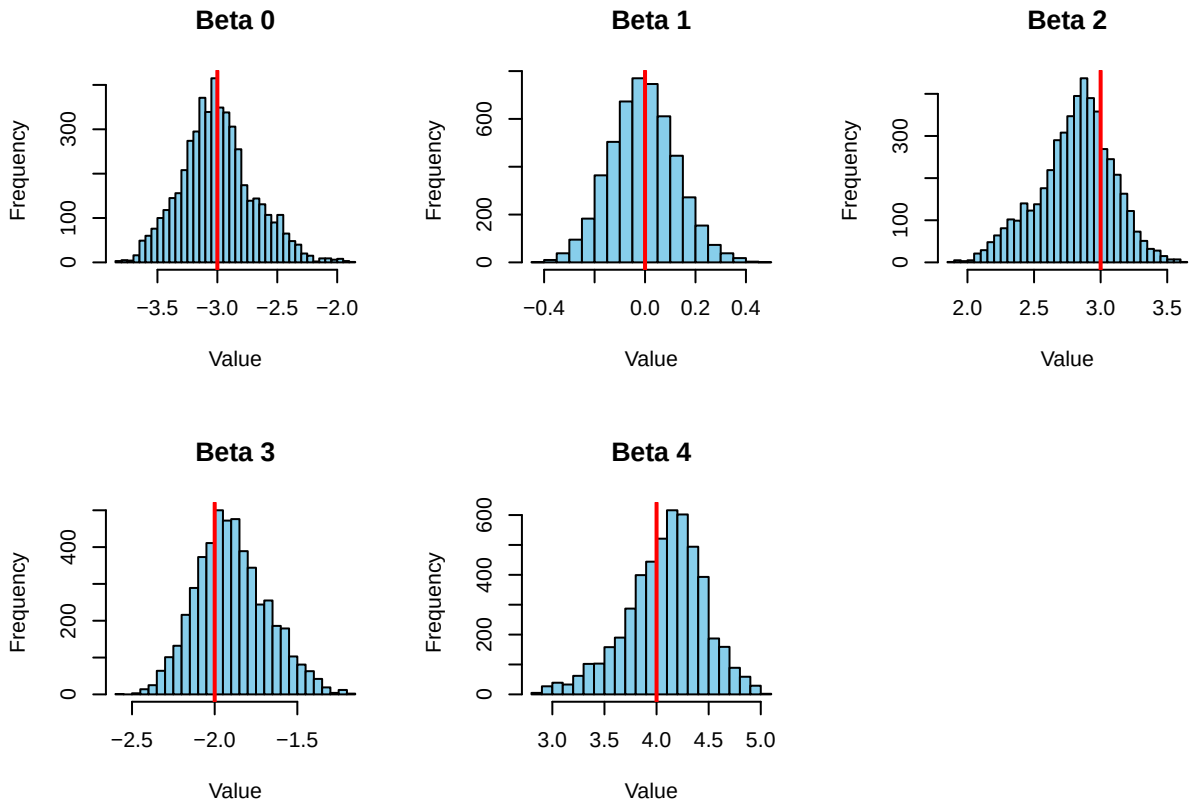
We will be using packages that can give us a truncated normal instead of doing extra work of creating our own sampling scheme for truncated normal.

```r
library(truncnorm)
```

```
## Warning: package 'truncnorm' was built under R version 4.5.3
```

```r
# Setup our variables and generate our starting stuff
set.seed(42)
n <- 500
p <- 5
beta_true <- c(-3, 0, 3, -2, 4)

X <- matrix(rnorm(n * p), n, p)
prob <- pnorm(X %*% beta_true)
Y <- rbinom(n, 1, prob)

# Gibbs sampler settings
n_iter <- 10000
burn_in <- 5000
```

```r
beta_samples <- matrix(0, n_iter, p)

# Precompute (X'X)^-1 and its Cholesky decomposition for later use
XtX_inv <- solve(t(X) %*% X)
chol_inv <- chol(XtX_inv)

# Running the steps detailed earlier
# Step 1: Initial value for beta is just a bunch of 0
current_beta <- rep(0, p)

for (s in 1:n_iter) {
  # Step 2: Sample Z_i | beta, X, Y
  mu_z <- X %*% current_beta
  Z <- ifelse(Y == 1,
              rtruncnorm(n, a = 0, b = Inf, mean = mu_z, sd = 1),
              rtruncnorm(n, a = -Inf, b = 0, mean = mu_z, sd = 1))

  # Step 3: Sample beta | Z, X
  # Mean of beta posterior from question part B: (X'X)^-1 X'Z
  mean_beta <- XtX_inv %*% t(X) %*% Z
  # Standard sample from Multivariate Normal using Cholesky decomp
  current_beta <- mean_beta + t(chol_inv) %*% rnorm(p)

  beta_samples[s, ] <- current_beta
}

# Remove burn-in
post_samples <- beta_samples[(burn_in + 1):n_iter, ]

# Visualization plot
par(mfrow = c(2, 3))
for (j in 1:p) {
  hist(post_samples[, j], main = paste("Beta", j-1),
       xlab = "Value", col = "skyblue", breaks = 30)
  abline(v = beta_true[j], col = "red", lwd = 2)
}
```



Seems like the sampling was not too bad compared to true β values.